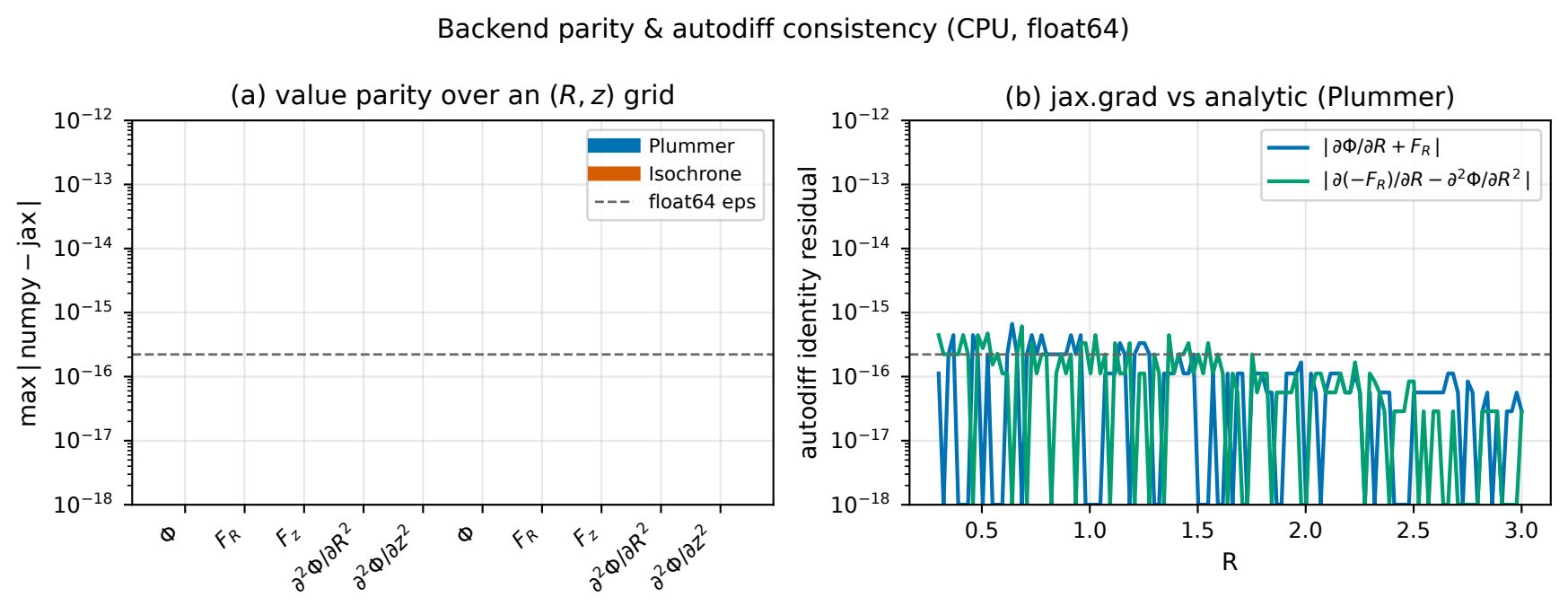
Backend parity & autodiff consistency (CPU, float64)
(a) value parity over an (R, z) grid
$\max|\text{numpy} - \text{jax}|$
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
10^{-18}
Plummer
Isochrone
float64 eps
Φ
F_R
F_z
$\partial^2\Phi/\partial R^2$
$\partial^2\Phi/\partial z^2$
Φ
F_R
F_z
$\partial^2\Phi/\partial R^2$
$\partial^2\Phi/\partial z^2$
(b) jax.grad vs analytic (Plummer)
autodiff identity residual
$|\,\partial\Phi/\partial R + F_R\,|$
$|\,\partial(-F_R)/\partial R - \partial^2\Phi/\partial R^2\,|$
0.5
1.0
1.5
2.0
2.5
3.0
R